3-13 Neuroprotection following cardiac arrest v.1

Outcome from cardiac arrest is determined by the severity of any supervening neurological or cardiac dysfunction / instability which results from poor vital organ perfusion. Following return of spontaneous circulation (ROSC), inability of the patient to obey commands indicates that neuroprotection techniques should be considered.

START

1. Prepare the cardiac arrest trolley for any further events.
2. Use positive pressure ventilation, aiming for:
   - SpO_2 > 94% and < 98%.
   - PCO_2 > 4.5 kPa and < 5.5 kPa.
3. Give sedation and neuromuscular blocking drugs to reduce thermogenesis from shivering.
4. Insert intra-arterial blood pressure monitoring. Consider vasopressor/inotrope to maintain systolic blood pressure, target SBP > 100 mmHg.
5. Obtain 12-lead ECG and discuss with cardiology if percutaneous coronary intervention is possible or appropriate.
6. Check blood glucose. Start glycaemic control therapies if above 10 $mmol.l^{-1}$.
7. Check core temperature. Target temperature is a constant temperature in the range of 32 – 36°C (precise target determined by local policy):
   - Temperature usually decreases without intervention in the immediate post-arrest period.
   - Start cooling strategies if indicated (Box A).
   - Avoid hyperthermia > 37.5°C.
8. Give antiepileptic drugs if seizures develop (Box B).
9. Plan further management in critical care area. Call for extra help as necessary.

Box A: COOLING STRATEGIES

Intravenous fluid bolus: if not contraindicated give 30 $ml.kg^{-1}$ of cold (4°C) non glucose-containing solutions
External: simple ice packs and/or wet towels; cooling blankets or pads; water or air circulating blankets; water circulating gel-coated pads
Internal: intravascular heat exchanger; cardiopulmonary bypass

Box B: DRUGS TO CONTROL/PREVENT SEIZURES

- Benzodiazepines or propofol are likely to be closest to hand in the operating theatre.
- Sodium valproate, levetiracetam, phenytoin or a barbiturate can also be used.

Box C: CRITICAL CHANGES

Cardiac arrest → **2-1**

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